Amir Hosein Maharati

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PERSONAL STATEMENT

A Medical Doctor and young researcher working in an Oncology Clinic where I assist in patient care and chemotherapy planning. My research focuses on non-coding RNAs as biomarkers in human tumors, and I'm deeply interested in bioinformatics, sequencing, and cancer biology. Beyond medicine, I enjoy theater and the arts, and I aspire to make a lasting difference in the lives of cancer patients

RESEARCH INTERESTS

Cancer biology	Mult omics	Bioinformatics
Cancer biomarkers	non-coding RNAs	Data analysis

EDUCATIONAL HISTORY

- **Doctor of medicine** 2018- 2025

Mashhad University of Medical Sciences

- *Cumulative GPA: 3.84/4 (18.12/20)*

Experience

Oncologist Assistant *June 2025-now*

Shafa Oncology and chemotherapy center (Dr. Allahyari)

Oncology Patient Management (over 1500 hours cancer patient management)

Treatment Planning & Prescription

Evidence-Based Practice

Collaboration

Undergraduate Research Assistant Jan 2020-2025

Medical Genetics Research Center (MGRC)

Supervisor: Dr. Mohammadreza Abbaszadedgan & Dr. Meysam Moghbeli

Developing a biobank for colorectal, gastric, esophageal squamous cell, papillary thyroid and ovarian cancer

Management of the colorectal projects at the research center

Developing collaborations with department heads for the purpose of data collection

Medical Intern Oct 2023-2025

Imam Reza and Ghaem hospital, Mashhad, Iran

Practicing medicine

Post-doctoral research fellow April 2025-2026

Endoscopic and minimally invasive research center, Ghaem Hospital, Mashhad, Iran

RESEARCH EXPERIENCE

◈ Major Publications

- **Maharati A**, Zanguei AS, Khalili-Tanha G, Moghbeli M. MicroRNAs as the critical regulators of tyrosine kinase inhibitors resistance in lung tumor cells. Cell Commun Signal. [**IF: 8.41**]

- Zohreh Rezaei, Kazem Dastjerdi, Abolghasem Allahyari, Soodabeh ShahidSales, Sahar Talebian, **Amirhosein Maharati**, Alireza Zangooie, Amir Sadra Zangouei, Farzad Sadri, Saman Sargazi. Plasma microRNA-195, −34c, and − 1246 as novel biomarkers for the diagnosis of trastuzumab-resistant HER2-positive breast cancer patients. Toxicology and Applied Pharmacology. [**IF: 4.4**]

- **Amirhosein Maharati,** Negin Taghehchian, Fatemeh Taghavinia, Alireza Golshan, Azadeh Aarabi, Mohammad Reza Abbaszadegan, Meysam Moghbeli.LINC01322 may serve as a potential diagnostic marker for advanced stage tumors in renal cell carcinoma patients eligible for total nephrectomy. Biochemistry and Biophysics Reports. **[IF: 2.2]**

- **Maharati A**, Moghbeli M. Long non-coding RNAs as the critical regulators of PI3K/AKT, TGF-β, and MAPK signaling pathways during breast tumor progression. *J Transl Med*. [**IF: 8.48**]

- **Maharati A**, Moghbeli M. PI3K/AKT signaling pathway as a critical regulator of epithelial-mesenchymal transition in colorectal tumor cells. *Cell Commun Signal*. [**IF: 8.41**]

- **Maharati A**, Samsami Y, Latifi H, Tolue Ghasaban F, Moghbeli M. Role of the long non-coding RNAs in regulation of Gemcitabine response in tumor cells. *Cancer Cell Int*. [**IF: 6.4**]

- **Maharati A**, Moghbeli M. Forkhead box proteins as the critical regulators of cisplatin response in tumor cells. *Eur J Pharmacol*. [**IF: 5.1**]

- Iman Akhlaghipour, Arya Nasimi Shad, Vahid Reza Askari, **Amirhosein Maharati**, Vafa Baradaran Rahimi. How caffeic acid and its derivatives combat diabetes and its complications: A systematic review. Journal of Functional Foods. [**IF: 5.2**]

- **Maharati A**, Akhlaghipour I, Taghehchian N, Farshchian Yazdi Z, Moghbeli M. Role of microRNA-494 in tumor progression. Am J Transl Res. [**IF: 3.9**]

- **Amirhosein Maharati,** Yasamin Rajabloo, Meysam Moghbeli. Molecular mechanisms of mTOR-mediated cisplatin response in tumor cells. Heliyon. **[IF: 3.6]**

- **Maharati A**, Moghbeli M. Role of microRNAs in regulation of doxorubicin and paclitaxel responses in lung tumor cells. *Cell Div*. [**IF: 2.8**]

- **Maharati A**, Tolue Ghasaban F, Akhlaghipour I, Taghehchian N, Zangouei AS, Moghbeli M. MicroRNA-495: a therapeutic and diagnostic tumor marker. *J Mol Histol*. [**IF: 3.1**]
- **Amirhosein Maharati**, Meysam Moghbeli. Role of microRNA-505 during tumor progression and metastasis. Pathology-Research and Practice. [**IF: 3.2**]

- Alsadat Mahmoudian R, **Amirhosein M**, Mahmoudian P, et al. The therapeutic potential value of Cancer-testis antigens in immunotherapy of gastric cancer. Gene. [**IF: 3.9**] [co-first author]

- Allahyari A, **Maharati A**, Jafari-Nozad AM, Zangooie A. Prostate cancer, chronic myelogenous leukemia and multiple myeloma in a single patient: a case report and review of the literature. *J Med Case Rep*. [co-respond author]

- Tolue Ghasaban F, **Maharati A**, Akhlaghipour I, Moghbeli M. MicroRNAs as the critical regulators of autophagy-mediated cisplatin response in tumor cells. *Cancer Cell Int*. [**IF: 6.4**]

- Taghehchian N, **Maharati A**, Akhlaghipour I, Zangouei AS, Moghbeli M. PRC2 mediated KLF2 down regulation: a therapeutic and diagnostic axis during tumor progression. *Cancer Cell Int*. [**IF: 6.4**]

- Tolue Ghasaban F, **Maharati A**, Zangouei AS, Zangooie A, Moghbeli M. MicroRNAs as the pivotal regulators of cisplatin resistance in head and neck cancers. *Cancer Cell Int*. [**IF: 6.4**]

- Fanoodi A, **Maharati A**, Akhlaghipour I, Rahimi HR, Moghbeli M. MicroRNAs as the critical regulators of tumor angiogenesis in liver cancer. *Pathol Res Pract*. [**IF: 3.3**]

- Moghbeli M, Taghehchian N, Akhlaghipour I, Samsami Y, **Maharati A**. Role of forkhead box proteins in regulation of doxorubicin and paclitaxel responses in tumor cells: A comprehensive review. *Int J Biol Macromol*. [**IF: 8.0**]

- Arya Nasimi Shad, Ali Fanoodi, **Amirhosein Maharati**, Iman Akhlaghipour, Amir Reza Bina, Ehsan Saburi, Fatemeh Forouzanfar, Meysam Moghbeli. Role of microRNAs in tumor progression by regulation of kinesin motor proteins. International journal of biological macromolecules. [**IF: 8.5**]

- Taghehchian N, Samsami Y, **Maharati A**, Zangouei AS, Boroumand-Noughabi S, Moghbeli M. Molecular biology of microRNA-342 during tumor progression and invasion. *Pathol Res Pract*. [**IF: 3.3**]

- Iman Akhlaghipour, Negin Taghehchian, Amir Sadra Zangouei, Amirhosein **Maharati**, Reihaneh Alsadat Mahmoudian, Ehsan Saburi, Meysam Moghbeli. MicroRNA-377: A therapeutic and diagnostic tumor marker. *Int J Biol Macromol*. [**IF: 8.0**]

- Nasimi Shad A, Fanoodi A, **Maharati A**, Akhlaghipour I, Moghbeli M. Molecular mechanisms of microRNA-301a during tumor progression and metastasis. *Pathol Res Pract*. [**IF: 3.3**]

- Malihe Lotfi, **Amirhosein Maharati,** Amir Abbas Hamidi, Negin Taghehchian, Meysam Moghbeli. MicroRNA-532 as a probable diagnostic and therapeutic marker in cancer patients. Mutation Research-Fundamental and Molecular Mechanisms of Mutagenesis. [**IF: 1.9**]

- Faezeh Tolue Ghasaban, Iman Akhlaghipour, Negin Taghehchian, **Amirhosein Maharati,** Bahram Memar, Meysam Moghbeli. MicroRNA-185: A non-invasive diagnostic and prognostic tumor marker. Process Biochemistry. [**IF: 4.8**]

- Alireza Zangooie, Zahra Moradi, Reza Asgari, **Amirhosein Maharati**, Zahra Salehi, Abolghasem Allahyari. Essential thrombocythemia as an initial presentation of polyneuropathy, organomegaly, endocrinopathy, M-protein, skin changes syndrome with complete response to the bortezomib, cyclophosphamide, dexamethasone regimen: **a case report. [IF: 0.9]**

- **Maharati A,** Zangooie A, Sardarkhani Eydgahi M, Haghshenas H, Eghbali Z, Tayyebi Meibodi N, Salehi Z, Allahyari A. Concurrent sigmoid colon adenocarcinoma presenting with cutaneous adult T-cell leukemia/lymphoma: a rare case report. Infect Agent Cancer. **[IF: 2.6]**
- Alireza Zangooie, Moein Piroozkhah, Zahra Ghasemi, Mobin Pirouzkhah, Maryam Sardarkhani Eydgahi, Reza Asgari, Malihe Saberafsharian, Zahra Sanei-Far**,** **Amirhosein Maharati**, Zahra Salehi, Abolghasem Allahyari. Discover Oncology. **[IF:3.5]**

◈ Under review research projects

- **GFPT2 as a critical regulator of CAFs-Tumor interaction in the TME of colorectal cancer (single cell)**
- **Critical regulators of CAFs-Tumor interaction through PI3K/ AKT in the TME of RCC (single cell)**
- **LRRC15⁺ myofibroblastic cancer-associated fibroblasts interaction with basal-squamous pancreatic ductal adenocarcinoma via c-MAF - TGF-β axis (single cell and spatial)**
- **LINC01789 as a malignant epithelial-associated biomarker in clear cell renal cell carcinoma (single cell and ATAC seq)**

OTHER EXPERIENCEs

- **Theater Director** Jan 2019-Mar 2022
  Mashhad University of Medical Sciences theater center, Mashhad, Iran
  - *performing 3 theaters as the director and 1 theater as an actor*
  - *training management, team work and communication skills*
- **Workout and fitness trainer** Oct 2022-now
  Iran bodybuilding and fitness federation, Tehran, Iran
  - *Level 3 Certificate in Coaching*

SOFTWARE DEVELOPMENT

- **HAVij: R package**
  High throughput association validation: a statistical package

AWARDS & ACHIEVEMENTS

- **Annual Top Researcher of University Award** Dec 2023
  Mashhad University of Medical Sciences

SKILLS & QUALIFICATIONS

- **Laboratory Skills**
  - DNA Extraction and PCR Techniques
  - RNA Extraction
  - Cell Culture
  - ELISA Techniques
  - MTT assay
  - Western Blot Technique

- Gram Staining
- Working with Laboratories Devices
- Microtome, Tissue Processing Device

- **Computer Skills**
  - Microsoft Office: Advanced
  - SPSS: Advanced
  - R Programming Language: Advanced
  - RNA-seq/ Single-cell Analysis: Advanced
  - Linux: Advanced

- **Single cell and spatial analysis**

  Seurat, Harmony, Monocle3, Signac, chromVAR, Cicero, epiAneufinder, DESeq2, edgeR, clusterProfiler, CellChat, LIANA, NicheNet, MultiNicheNet, CellPhoneDB, SCENIC, pySCENIC, AUCell, GENIE3, RcisTarget, RCTD, Squidpy, Scanpy, SingleR, trade seq, TERRA, mintflow, nichecompass

- **Language Skills**
  - Farsi: Fluent
  - Turkish: Fluent
  - English: Proficient

REFERENCES

- **Mohammadreza Abbaszadegan (Ph.D.)**
  Professor of Medical Genetics, Mashhad University of Medical Sciences
  Email: m.r.abbaszadegan@gmail.com
- **Meysam moghbeli (Ph.D.)**
  Associate Professor of Molecular Genetics, Mashhad University of Medical Sciences
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